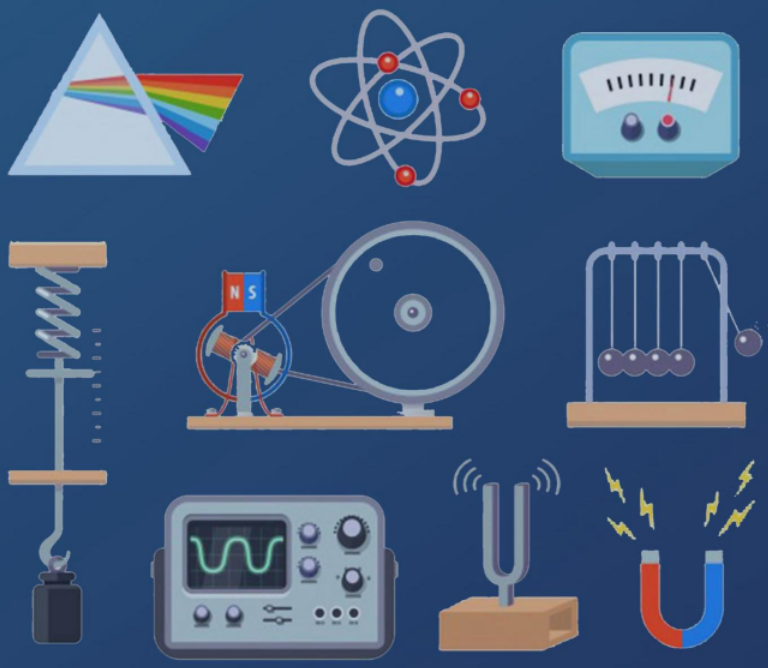
PHYSICS

Errors in Measurement



SHEET

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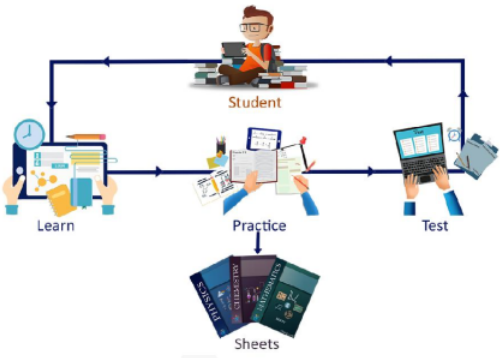
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Types of Errors	1			
2	A Question to Learn More			1	
3	Combination of Errors	2-10		2-4	17,18,25,26
4	Vernier Callipers	11-15	1,3	5-7	1
5	Screw Gauge	16,17	4,8	8-11	2-5,14-16
6	Significant Digits	18-20	5-7	12-14	19-21
7	Miscellaneous Problems		2	15	6-10,22,23
8	After Full Chapter	21		16	11-13,24

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Types of Errors

Q1. The mean time period of a simple pendulum is 1.92 s. Mean absolute error in the time period is 0.05 s. To express the maximum estimate of error, the time period should be written as–

(A) $T = (1.92 \pm 0.01)$ s

(B) $T = (1.92 \pm 0.025)$ s

(C) $T = (1.92 \pm 0.05)$ s

(D) $T = (1.92 \pm 0.10)$ s

Combination of Errors

Q2. A quantity is represented by $X = M^a L^b T^c$. The percentage error in measurement of M, L and T are α%, β % and γ % respectively. The percentage error in X would be :

(A) $(\alpha a + \beta b + \gamma c)$ %

(B) $(\alpha a - \beta b + \gamma c)$ %

(C) $(\alpha a - \beta b - \gamma c)$ %

(D) None of these

Q3. An experiment measures quantities a, b and c and X is calculated from $X = ab^2/c^3$. If the percentage error in a, b and c are $\pm 1\%$, $\pm 3\%$ and $\pm 2\%$ respectively, the percentage error in X will be :-

(A) $\pm 13\%$ (B) $\pm 7\%$

(C) $\pm 4\%$ (D) $\pm 1\%$

Q4. What is the fractional error in g calculated from $T = 2\pi\sqrt{\ell/g}$? Given that fractional errors in T and ℓ are $\pm x$ and $\pm y$ respectively.

(A) $x + y$ (B) $x - y$

(C) $2x + y$ (D) $2x - y$

Q5. A thin copper wire of length ℓ metre increases in length by 2% when heated through 10°C. What is the percentage increase in area when a square copper sheet of length ℓ metre is heated through 10°C ?

(A) 4%

(B) 8%

(C) 16%

(D) None of the above.

Q6. The resistance is $R = \frac{V}{I}$,

where $V = 100 \pm 5$ Volts

and $I = 10 \pm 0.2$ amperes.

What is the total error in R ?

(A) 5 % (B) 7 %

(C) 5.2 % (D) $\left(\frac{5}{2}\right)$ %

Q7. The length, breadth and thickness of a strip are (10.0 ± 0.1) cm, (1.00 ± 0.01)cm and (0.100 ± 0.001)cm respectively. The most probable error in its volume will be :

(A) ± 0.03 cm^3

(B) ± 0.111 cm^3

(C) ± 0.012 cm^3

(D) None of these

Q8. The percentage error in the measurement of mass and speed are 1% and 2% respectively. What is the percentage error in kinetic energy :-

(A) 5% (B) 2.5%

(C) 3% (D) 1.5%

Q9. If error in measuring diameter of a circle is 4 %, the error in circumference of the circle would be :-

(A) 2% (B) 8%

(C) 4% (D) 1%

Q10. The external and internal radius of a hollow cylinder are measured to be (4.23 ± 0.01) cm and (3.89 ± 0.01)cm. The thickness of the wall of the cylinder is :-

(A) (0.34 ± 0.02) cm

(B) (0.17 ± 0.02) cm

(C) (0.17 ± 0.01) cm

(D) (0.34 ± 0.01) cm

Vernier Caliper

Q11. In a vernier calliper, N divisions of vernier scale coincide with (N – 1) divisions of main scale (in which 1 division represents 1mm). The least count of the instrument in cm should be :

(A) N (B) N – 1

(C) $\frac{1}{10N}$ (D) $\frac{1}{N-1}$

Q12. One centimetre on the main scale of vernier callipers is divided into ten equal parts. If 10 divisions of vernier scale coincide with 8 small divisions of the main scale, the least count of the callipers is -

(A) 0.01 cm (B) 0.02 cm

(C) 0.05 cm (D) 0.005 cm

Q13. The zero error in a Vernier callipers is said to be positive when -

(A) zero of vernier scale is towards the right of the zero of main scale

(B) zero of vernier scale coincides with the zero of main scale

(C) zero of vernier scale is towards the left of the zero of main scale

(D) the vernier scale is not visible clearly.

Q14. Using upper pair of jaws of Vernier Callipers, we can measure :

(A) the depth of a cylindrical vessel

(B) the outer diameter of a vessel

(C) the inner diameter of a vessel

(D) the thickness of a thin wire

Q15. The zero error in a Vernier Callipers is said to be negative when -

(A) zero of Vernier Scale is towards right of the zero of main scale

(B) zero of Vernier scale coincides with the zero of main scale.

(C) zero of Vernier scale is towards left of the zero of main scale

(D) the Vernier scale is not visible clearly.

Screw gauge

Q16. A screw gauge has 1.0mm pitch and 200 divisions on the circular scale. The least count of the instrument is -

(A) 5×10^{-3} mm

(B) 5×10^{-4} mm

(C) 5×10^{-2} mm

(D) 5×10^{-5} mm

Q17. The circular scale of a screw gauge has 200 divisions. When it is given 4 complete rotations, it moves through 2mm. The least count of the screw gauge is -

(A) 0.25×10^{-2} cm

(B) 0.25×10^{-3} cm

(C) 0.001 cm

(D) 0.001 mm

Significant digits

Q18. How many significant figures are given in the following quantities ?

(a) 343 g (b) 2.20

(c) 1.103 N (d) 0.4142 s

(e) 0.0145 m (f) 1.0080 V

(g) 9.1×10^4 km (h) 1.124×10^{-3}V

(A) 24 (B) 26

(C) 28 (D)30

Q19. Which of the following option is correct for performing following operations :

(i) $703 + 7 + 0.66$ (ii) 2.21×0.3

(iii) 12.4×84 (iv) $14.28 / 0.714$

(A) 711, 0.7, 1.0×10^3, 20.0

(B) 0.7, 711, 1×10^3, 20

(C) 711, 20, 1×10^3, 0.7

(D) 711, 1×10^3, 0.7, 20

Q20. Solve with due regard to significant digits -

(i) $\sqrt{6.5 - 6.32}$ (ii) $\dfrac{2.91 \times 0.3842}{0.080}$

(A) 0.4, 14 (B) 0.4,13

(C) 0.3,14 (D) 0.3,13

After Full Chapter

Q21. The time period of a simple pendulum is given by $T = 2\pi\sqrt{\dfrac{\ell}{g}}$. In an experiment, the length of the pendulum is increased by 1%, while the acceleration due to gravity is also increased by 2%. The time period will –

Note :– It is talking of change & not error

(A) increase by 0.5%

(B) decrease by 0.5%

(C) increase by 1%

(D) decrease by 1%

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The nth division of main scale coincides with $(n + 1)^{th}$ division of vernier scale. Given one main scale division is equal to 'a' units. The least count of vernier is -

(A) $\dfrac{n}{a+1}$ (B) $\dfrac{a}{n+1}$

(C) an (D) $\dfrac{a}{n}$

Q2. In the Searle's experiment, after every step of loading , why should we wait for two minutes before taking the readings?

(A) So that the wire can have its desired change in length.

(B) So that the wire can attain room temperature.

(C) So that vertical oscillations can get subsided.

(D) So that the wire has no change in its radius.

SUBJECTIVE TYPE

Q3. One centimetre on the main scale of vernier callipers is divided into ten equal parts. If 20 divisions of vernier scale coincide with 19 small divisions of the main scale then what will be the least count of the callipers.

Q4. If the number of divisions on the circular scale is 100 and number of full rotations given to screw is 8 and distance moved by the screw is 4 mm, then what will be least count of the screw gauge.

Q5. Calculate area enclosed by a circle of diameter 1.06 m to correct number of significant figures.

Q6. Subtract 2.5×10^4 from 3.9×10^5 and give the answer to correct number of significant figures.

Q7. The original length of wire is (153.7 ± 0.6) cm. It is stretched to (155.3 ± 0.2) cm. Calculate the elongation in the wire with error limits.

NUMERICAL TYPE

Q8. The pitch of a screw gauge is 0.5 mm and there are 50 divisions on the circular scale. In measuring the thickness of a metal plate, there are five divisions on the pitch scale (or main scale) and thirty fourth division coincides with the reference line. Calculate the thickness of the metal plate in (mm).

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The period of oscillation of a simple pendulum in the experiment is recorded as 2.63s, 2.56s, 2.42s, 2.71s and 2.80s respectively. The average absolute error is -

(A) 0.1s (B) 0.11s

(C) 0.01s (D) 1.0s

Q2. The length of a cylinder is measured with a metre rod having least count 0.1 cm. Its diameter is measured with vernier callipers having least count 0.01 cm. Given the length is 5.0 cm. and radius is 2.00 cm. The percentage error in the calculated value of volume will be –

(A) 2% (B) 1%

(C) 2.5% (D) 4%

Q3. The heat generated in a circuit is dependent upon the resistance, current and time for which the current is flown. If the error in measuring the above are as 1%, 2% and 1% the maximum error in measuring heat will be

(A) 2% (B) 3%

(C) 6% (D) 1%

Q4. A wire has a mass (0.3 ± 0.003)g, radius (0.5 ± 0.005) mm and length (6 ± 0.06) cm. The maximum percentage error in the measurement of its density is –

(A) 1 (B) 2

(C) 3 (D) 4

Q5. A vernier callipers has 20 divisions on the vernier scale which coincide with 19 divisions on the main scale. The least count of the instrument is 0.1 mm. The main scale divisions are of

(A) 0.5 mm (B) 1 mm

(C) 2 mm (D) 1/4 mm

Q6. The vernier constant of a vernier callipers is 0.1mm and it has a positive zero error of 0.04cm. While measuring diameter of a rod, the main scale reading is 1.2 cm and 5th vernier division is coinciding with any scale division. The correct diameter of the rod is-

(A) 1.21 cm (B) 1.21 mm

(C) 1.29 mm (D) 1.29 cm

Q7. 1 cm of main scale of a vernier calliper is divided into 10 divisions. The least count of the callipers is 0.05 cm, then the varnier scale must have :

(A) 10 divisions (B) 20 divisions

(C) 25 divisions (D) 50 divisions

Q8. A screw gauge gives the following reading when used to measure the diameter of a wire. Main scale reading = 0 mm, circular scale reading = 52 divisions. Given that 1mm on main scale corresponds to 100 divisions of the circular scale. The diameter of the wire from the above data is -

(A) 0.052 cm (B) 0.026 cm

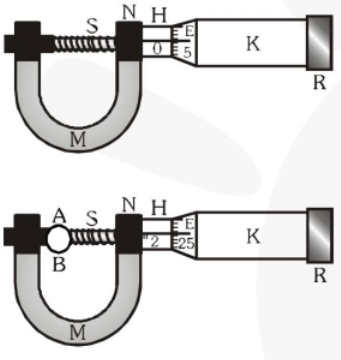
(C) 0.005 cm (D) 0.52cm

Q9. When a screw gauge is completely closed, zero of circular scale is 6 divisions below the reference line of graduation. If least count of screw gauge is 0.001 cm, the zero correction is -

(A) –0.006 cm (B) +0.006 cm

(C) –0.006 mm (D) +0.006 mm

Q10. The circular scale of a screw gauge has 50 divisions and pitch of 0.5mm. Find the diameter of sphere. Main scale reading is 2



(A) 1.20 mm (B) 1.25 mm

(C) 2.207 mm (D) 2.25 mm

Q11. A student measured the diameter of a wire using a screw gauge with least count 0.001 cm and listed the measurements. The correct measurement is -

(A) 5.3 cm (B) 5.32 cm

(C) 5.320 cm (D) 5.3200 cm

Q12. Least count of a vernier callipers is 0.01cm. Using this, the diameter of a sphere is measured as 1.95cm. Radius of the sphere to the correct significant figure will be :

(A) 0.98 cm (B) 0.975 cm

(C) 1.0 cm (D) 1 cm

Q13. Measure of two quantities along with the precision of respective measuring instrument is :-

A = 2.5 m/s ± 0.5 m/sB = 0.10s ± 0.01 s

The value of AB will be :-

(A) (0.25 ± 0.08) m

(B) (0.25 ± 0.5) m

(C) (0.25 ± 0.05) m

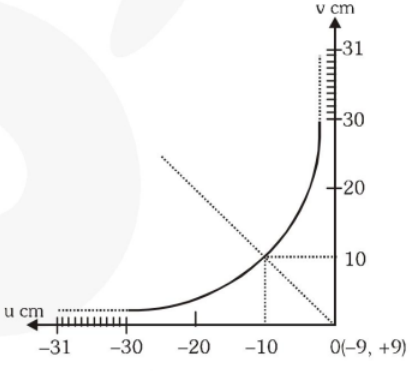
(D) (0.25 ± 0.135) m

Q14. The edge of a cube is $a = 1.2 \times 10^{-2}$ m. Then its volume will be recorded as :

(A) 1.7×10^{-6} m^3 (B) 1.70×10^{-6} m^3

(C) 1.70×10^{-7} m^3 (D) 1.78×10^{-6} m^3

Q15. Graph of position of image vs position of point object from a convex lens is shown. Then, focal length of the lens is -



(A) 0.50 ± 0.05 cm (B) 0.50 ± 0.10 cm

(C) 5.00 ± 0.05 cm (D) 5.00 ± 0.10 cm

Q16. An experiment is performed to find the refractive index of glass using a travelling microscope. In this experiment distances are measured by

(A) a vernier scale provided on the microscope

(B) a stanard laboratory scale

(C) a meter scale provided on the microscope

(D) a screw gauge provided on the microscope

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The least count of vernier callipers is 0.1mm. The main scale reading before the zero of the vernier scale is 10 and the zero th division of the vernier scale coincides with the main scale division. Given that each main scale division is 1 mm, the measured value should be expressed.

(A) 1 cm (B) 2 cm
(C) 0.5 cm (D) 0.1 cm

Q2. Two full turns of the circular scale of gauge cover a diastance of 1 mm on scale. The total number of divisions on circular scale is 50. Further, it is found that screw gauge has a zero error of -0.03 mm. While measuring the diameter of a thin wire a student notes the main scale reading of 3 mm and the number of circular scale division in line, with the main scale as 35. The diameter of the wire is-

(A) 3.32 mm (B) 3.73 mm
(C) 3.67 mm (D) 3.38 mm

Q3. When the gap is closed without placing any object in the screw gauge whose least count is 0.005 mm, the 5th division on its circular scale coincides with the reference line on main scale and when a small sphere is placed reading on main scale advances by 4 divisions, whereas circular scale reading advances by five times to the corresponding reading when no object was placed. There are 200 divisions on the circular scale. The radius of the sphere is -

(A) 4.10 mm (B) 4.05 mm
(C) 2.10 mm (D) 2.05 mm

Q4. While measuring diameter of a wire using a screw gauge the main scale reading is 7mm and zero of circular scale is 35 divisions above the reference line. If the screw gauge has a zero error of –0.003 cm, the correct diameter of the wire is

(given least count = 0.001cm)

(A) 0.735cm (B) 0.732cm
(C) 0.738 cm (D) 7.38 cm

Q5. On measuring diameter of a wire with help of screw gauge, main scale reading is 1 mm and 6th division of circular scale lying over reference line. On measuring zero error, it is found that zero of circular scale has advanced from reference line by 3 divisions on circular scale, then corrected diameter is-

(A) 1.09 mm (B) 1.06 mm
(C) 1.03 mm (D) 1.60 mm

Q6. A student obtained following observations in an experiment of meter bridge to find unknown resistance of given wire :

S.No.	R	ℓ	100 – ℓ	$S=\left(\frac{100-\ell}{\ell}\right)R$
1	2Ω	43	57	2.65
2	3Ω	52	48	2.77
3	4Ω	58	42	2.89
4	5Ω	69	31	2.25

The most accurate value of unknown resistance will be :

(A) 2.65 Ω (B) 2.77 Ω
(C) 2.89 Ω (D) 2.25 Ω

Q7. While measuring the speed of sound by performing a resonance column experiment, a student get's the first resonance condition at a column length of 18 cm during winter. Repeating the same experiment during summer, she measures the column length to be x cm for the second resonance. Then-

(A) x > 54 (B) 54 > x > 36
(C) 36 > x > 18 (D) 18 > x

Q8. A student performs an experiment for determination of $\left(g=\frac{4\pi^2\ell}{T^2}\right)$, $\ell \approx 1$m, and he commits an error of $\Delta\ell$. For T he takes the time of n oscillations with the stop watch of least count ΔT and he commits a human error of 0.1 s. For which of the following data, the measurement of g will be most accurate ?

(A) $\Delta L = 0.5, \Delta T = 0.1, n = 20$

(B) $\Delta L = 0.5, \Delta T = 0.1, n = 50$

(C) $\Delta L = 0.5, \Delta T = 0.01, n = 20$

(D) $\Delta L = 0.1, \Delta T = 0.05, n = 50$

Q9. A student performs an experiment to determine the Young's modulus of a wire, exactly 2m long, by Searle's method. In a particular reading, the student measures the extension in the length of the wire to be 0.8 mm with an uncertainty of ± 0.05 mm at a load of exactly 1.0 kg. The student also measures the diameter of the wire to be 0.4 mm with an uncertainty of ± 0.01 mm. Take $g = 9.8$ m/s^2 (exact). The Young's modulus obtained from the reading is –

(A) $(2.0 \pm 0.3) \times 10^{11}$ N/m^2

(B) $(2.0 \pm 0.2) \times 10^{11}$ N/m^2

(C) $(2.0 \pm 0.1) \times 10^{11}$ N/m^2

(D) $(2.0 \pm 0.05) \times 10^{11}$ N/m^2

Q10. A student measured the length of a rod and wrote it as 3.50 cm. which instrument did the use to measure it?

(A) A screw gauge having 100 divisions in the circular scale and pitch as 1 mm.

(B) A screw gauge having 50 divisions in the circular scale and pitch as 1 mm.

(C) A meter scale

(D) A vernier calliper where the 10 divisions in vernier scale matches with 9 divisions in main scale and main scale has 10 divisions in 1 cm.

Q11. Students I, II and III perform an experiment for measuring the acceleration due to gravity (g) using pendulum. They use different lengths of the pendulum and / or record time for different number of oscillations. The observations are shown in the table.

Least count for length = 0.1 cm

Least count for time = 0.1 s

Student	Length of the Pendulum (cm)	Number of oscillations (n)	Total time for (n) oscillations (s)	Time period (s)
I	64.0	8	128.0	16.0
II	64.0	4	64.0	16.0
III	20.0	4	36.0	9.0

If E_I, E_{II} and E_{III} are the percentage error in g, i.e., $\left(\frac{\Delta g}{g}\times 100\right)$ for students I, II and III, respectively.

(A) $E_I = 0$

(B) E_I is minimum

(C) $E_I = E_{II}$

(D) E_{II} is maximum

Q12. A student performs an experiment for determination of $g\left(=\frac{4\pi^2 l}{T^2}\right)$ $l \approx 1$m and he commits an error of Δl. For the experiment takes the time of n oscillations with the stop watch of least count ΔT and he commits a human error of 0.1sec. For which of the following data, the measurement of g will be most accurate?

	Δl	ΔT	n	Amplitude of oscillation
(A)	5 mm	0.2 sec	10	5 mm
(B)	5 mm	0.2 sec	20	5 mm
(C)	5 mm	0.1 sec	20	1 mm
(D)	1 mm	0.1 sec	50	1 mm

Q13. On the basis of detail given about two measuring instruments, select the correct statement.

(i) Vernier callipers having main scale division = 0.05 cm and Vernier scale division $= \frac{2.45}{50}$ cm.

(ii) Screw gaugae having pitch 0.5 mm and its circular scale division measures 0.01 mm.

(A) Both the instruments have same least count.

(B) Least count of screw gauge is lesser than that of vernier callipers.

(C) Both the instruments have same least count but screw gauge is more precise.

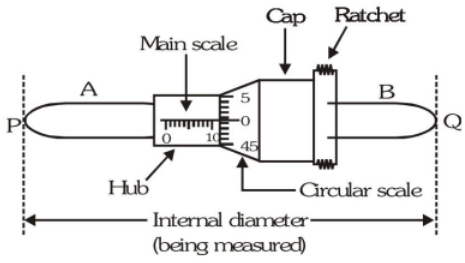
(D) Both the instruments have different least count and screw gauge is more precise.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 14 to 16

Internal micrometer is a measuring instrument used to measure internal diameter (ID) of a large cylinder bore with high accuracy. Construction is shown in figure. There is one fixed rod B (to the right in figure) and one moving rod A (to the left in figure). It is based on the principle of advancement of a screw when it is rotated in a nut with internal threads. Main scale reading can be directly seen on the hub which is fixed with respect to rod B. When the cap is rotated, rod A moves in or out depending on direction of rotation. The circular scale reading is seen by checking which division of circular scale coincides with the reference line.

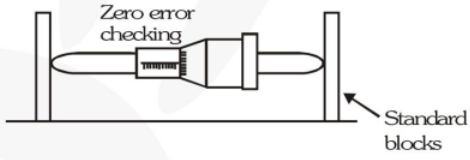


This is to be multiplied by LC to get circular scale reading.

Least count = value of 1 circular scale division

$$= \frac{\text{pitch}}{\text{number of division on circular scale}}$$

Length of rod A is chosen to match the ID (PQ) to be measured. Zero error is checked by taking reading between standard blocks fixed at nominal value of ID to be measured. Zero error is positive if cap end is on the right side of the main scale and negative it is on the left side.



Q14. In an internal micrometer, main scale division is of 0.5 mm and there are 50 divisions in circular scale. The least count of the instrument is -

(A) 0.005 mm

(B) 0.001 mm

(C) 0.05 mm

(D) 0.01 mm

Q15. In the above instrument, while measuring an internal diameter. ID is set of 321 mm with no zero error. If cap end is after 7th division and 17th division of main scale coincides with the reference line, the ID is -

(A) 321. 717 mm

(B) 321.87 mm

(C) 328.17 mm

(D) 324.67 mm

Q16. During zero setting of the above instrument, the end of the cap is on left side of the zero of main scale (i.e. zero of main scale is not visible) and 41st division of circular scale coincides with the reference line, the zero error is -

(A) –0.09 mm (B) +0.41 mm

(C) –0.41 mm (D) +0.09 mm

SUBJECTIVE TYPE

Q17. The time period of oscillation of a simple pendulum is given by $T = 2\pi\sqrt{\ell / g}$. The length of the pendulum is measured as $\ell = 10 \pm 0.1$ cm and the time period as $T = 0.5 \pm 0.02$ s. Determine percentage error in the value of g.

Q18. A body travels uniformly a distance of (13.8 ± 0.2)m in time (4.0 ± 0.3) sec. Calculate its velocity.

Q19. The pitch of a screw gauge is 1 mm and there are 100 divisions on the circular scale. While measuring the diameter of a wire, the linear scale reads 1 mm and 47th division on the circular scale coincides with the reference line. The length of the wire is 5.6 cm. Find the curved surface area (in cm^2) of the wire in appropriate number of significant figures.

Q20. The edge of a cube is measured using a vernier calliper. (9 divisions of the main scale is equal to 10 divisions of vernier scale and 1 main scale division is 1 mm). The main scale division reading is 10 and 1 division of vernier scale was found to be coinciding with the main scale. The mass of the cube is 2.736 g. Calculate the density in g/cm^3 upto correct significant figures.

Q21. 5.74 gm of a substance occupies a volume of 1.2 cm^3. Calculate its density with due regard for significant figures.

Q22. In a Searle's experiment, the diameter of the wire as measured by a screw gauge of least count 0.001 cm is 0.050 cm. The length, measured by a scale of least count 0.1 cm, is 110.0 cm. When a weight of 50N is suspended from the wire, the extension is measured to be 0.125 cm by a micrometer of least count 0.001 cm. Find the maximum error in the measurement of Young's modulus of the material of the wire from these data.

Q23. Two resistance R_1 and R_2 are connected in (i) series and (ii) parallel.

What is the equivalent resistance with limit of possible percentage error in each case of $R_1 = 5.0 \pm 0.2\ \Omega$ and $R_2 = 10.0 \pm 0.1\ \Omega$.

Q24. A glass prism of angle A = 60° gives minimum angle of deviation θ = 30° with the max. error of 1° when a beam of parallel light passed through the prism during an experiment. Find the permissible error in the measurement of refractive index μ of the material of the prism.

NUMERICAL TYPE

Q25. A physical quantity P is related to four observables A, B, C and D as $P = 4\pi^2 A^3 B^2 / (\sqrt{C}\, D)$. The percentage error of the measurement in A, B, C and D are 1%, 3% and 2%, 4% respectively. Determine the percentage error & absolute error in the quantity P. Value of P is calculated 3.763.

Q26. A wire of length $\ell = 6 \pm 0.06$ cm and radius $r = 0.5 \pm 0.005$ cm and mass $m = 0.3 \pm 0.003$ gm. Maximum percentage error in density is -

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	A	A	C	A	B	A	A	C	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	B	A	C	C	A	B	C	A	A
21									
B									

Ex-1A

1. B **2.** A,B,C **3.** 0.005 cm **4.** 0.005 mm **5.** 0.882 m^2

6. 3.6×10^5 (2 SF) **7.** (1.6 ± 0.8) cm **8.** 2.84

Ex - 2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	C	C	D	C	A	B	A	A	A
11	**12**	**13**	**14**	**15**	**16**				
C	A	A	A	C	A				

Ex-2A

1. A **2.** D **3.** D **4.** 3 **5.** 1

6. B **7.** A **8.** D **9.** B **10.** 4

11. B **12.** D **13.** A **14.** D **15.** D

16. A **17.** 9 **18.** $v = (3.45 \pm 0.3)$ m/s **19.** 2.6 cm^2

20. 2.66 g/cm^3 **21.** [4.8 g/cm^3] **22.** 4.89% **23.** [$R_s = 15\Omega \pm 2\%$, $R_p = 3.3\ \Omega \pm 3\%$]

24. $5\pi/18\%$ **25.** 14, 0.526 **26.** 4